

2. Precise typing of pattern matching (I)

Consider the following pattern matching expression

`match e with  $p_1 \rightarrow e_1 \mid p_2 \rightarrow e_2$`

where patterns are defined as follows:

$p ::= x \mid (p, p) \mid p \mid p \mid p \wedge p$

If we interpret types as set of values

$t = \{v \mid v \text{ is a value of type } t\}$

then **the set of all values that match a pattern is a type**

$\{p\} = \{v \mid v \text{ is a value that matches } p\}$

accepted
Type of p

$$\begin{aligned}\{x\} &= \mathbb{1} \\ \{(p_1, p_2)\} &= \{p_1\} \times \{p_2\} \\ \{p_1 \mid p_2\} &= \{p_1\} \vee \{p_2\} \\ \{p_1 \wedge p_2\} &= \{p_1\} \wedge \{p_2\}\end{aligned}$$